

Unit

1

Lesson

4

Spark The Challenge!

Ramp up the Fun! DIY Spiral Park

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Use ramps, shapes, and colors to build fun and safe play areas.
- Design and build a model of a spiral ramp.
- Try out different materials to see which ones are strong and safe.
- Count and compare objects used in construction projects.
- Use number lines, charts, and ordinal numbers to plan, organize, and sequence design elements.

Challenge questions of the lesson

1. What makes a spiral ramp better than a straight one for some kids?
2. How can you make sure your spiral ramp is strong and doesn't fall over?
3. How does the shape or color of a ramp help make it safer for all kids?

SEL Relation

- Planning and Organization (Self-Management): Students will practice planning the initial stages of their physical construction and organizing their materials and workspace.
- Problem-Solving (Self-Awareness & Responsible Decision-Making): Students will encounter challenges in translating their digital designs and will need to problem-solve how to represent them physically.
- Perseverance (Self-Management): Building can sometimes be tricky, and students will need to persevere through any difficulties they encounter.
- Spatial Awareness (Social Awareness - Understanding others' perspectives): Students will begin to think about how the layout of the playground will affect how others move and play within the space.

Engage



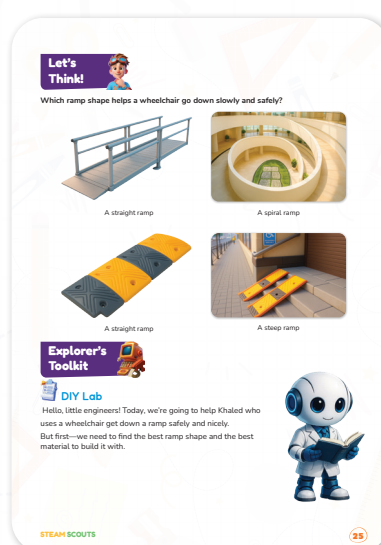
Teacher Role:

- **Facilitator:** Guide the discussion about the story, ensuring all students have a chance to share their thoughts and ideas about how to help Khaled.
- **Questioner:** Ask open-ended questions to encourage critical thinking and problem-solving related to ramp design and construction.
- **Motivator:** Build excitement for the upcoming building activity, emphasizing the real-world application of their ideas.
- **Resource Organizer:** Prepare the necessary materials (cardboard, tape, blocks) for the building activity that will likely follow.
- **Safety Monitor:** If the building activity proceeds, ensure students are using materials safely.

Expected From Students:

- **Active Listening:** Students will listen attentively to the story and understand the challenge of helping Khaled get to the slide.
- **Empathy:** Students will consider Khaled's perspective and the desire of Adam and Laila to find a solution.
- **Idea Generation:** Students will think about different ways to create a path that goes up, similar to the brainstorming in the story.
- **Participation in Discussion:** Students will share their ideas and respond to the suggestions made by the characters in the story and their classmates.
- **Problem-Solving:** Students will begin to think about the practicalities of building a ramp using the suggested materials.

Parts of the books included.



Story



1. Set the Scene with Excitement

“Today we’re continuing the story of Adam, Laila, and Khaled! Remember how Khaled couldn’t get to the slide? Let’s see what idea they came up with next!”

2. Read the Dialogue with Expression

- Use voices for each character to bring the story to life.
- Act out the parts with gestures (e.g., swirling finger for a spiral ramp).

3. Ask a Think-Aloud Question

“Have you ever seen a ramp that goes around and around? Why might that be a helpful idea?”

4. Connect to the Activity

“Just like in the story, you’re going to become engineers today and figure out how to build a spiral ramp that helps everyone get to the slide!”

Possible Strategies for Implementation:

1. Story Reenactment

- Let students act out the story with props or simple puppets.
- Pause at “Let’s get our materials and start building!” to launch the hands-on portion.

Engages students emotionally and builds ownership over the design challenge.

Let's Think!



Objective: Students will analyze and compare different ramp shapes and slopes to determine which type allows a wheelchair to go down slowly and safely. Through visual reasoning and guided discussion, students will build understanding of how design impacts accessibility and safety in playgrounds and public spaces.

Answer: Spiral ramp

A spiral ramp has a gentle, continuous slope that allows the wheelchair to descend gradually and more slowly than a steep or short straight ramp. It also often includes railings and turns that reduce speed naturally, making it safer for users with mobility needs.



Possible Strategies for implementation:

1. Visual Analysis Discussion

- Project the image or show it on paper and ask:
“What do you notice about each ramp?”
“Which one looks safe to go down slowly?”

Supports visual learners and encourages careful observation.

2. Think-Pair-Share

- Ask the question:
“Which ramp shape helps a wheelchair go down slowly and safely?”
- Let students think quietly, then discuss with a partner, and finally share with the class.

Builds critical thinking and communication skills.

3. Real-Life Connection

- Show or bring a toy wheelchair or small figure and use ramps made from cardboard to test different slopes (flat, spiral drawn, steep).
- Ask:
“Which one would you want to use if you were on wheels?”

Promotes empathy and real-world connection.

4. Anchor Chart: What Makes a Ramp Safe?

- After discussing, create a chart with students listing:
 - Gentle slope
 - Long path
 - Railings or sides
 - Not too steep

Gives a visual reference for future design decisions.

5. Prediction Check

- Before revealing or testing, ask students to make a prediction:
“Which ramp do you think is safest and why?”

Encourages reasoning and supports formative assessment.

Explorer's Toolkit



Teacher Role:

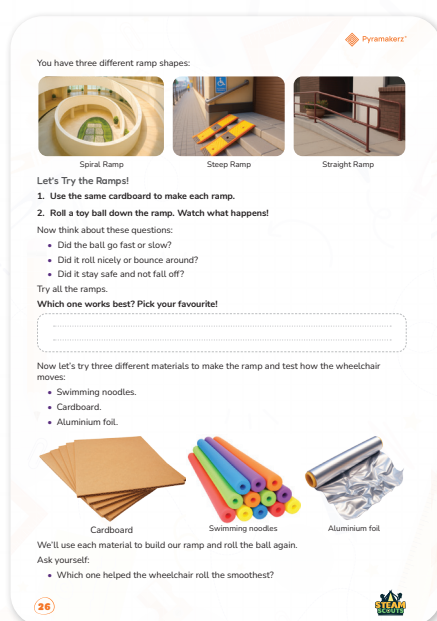
- **Introducer:** Introduce the activity, reminding students of the goal: to help Khaled get his wheelchair down a ramp safely and nicely. Emphasize that they will be testing different ramp shapes and materials.

- **Facilitator:** Guide students through the ramp building and testing process, ensuring they understand the instructions and are working safely.
- **Questioner:** Prompt students to observe carefully and think critically about how the ramp shape and material affect the movement of the toy ball (representing the wheelchair). Ask guiding questions during the exploration and reflection phases.
- **Resource Manager:** Ensure students have access to the cardboard, rulers, tape, and toy balls. If the second part of the activity is conducted, provide swimming noodles, cardboard, and aluminum foil.
- **Observer:** Observe students' experimental approaches, their ability to record observations, and their reasoning during the reflection.

Expected From Students:

- **Hands-on Participation:** Students will actively participate in building simple ramps out of cardboard.
- **Experimentation:** Students will roll a toy ball down their ramps and observe its movement.
- **Observation:** Students will carefully watch how the ball moves on different ramp shapes, noting its speed, smoothness, and stability.
- **Data Collection (Informal):** Students will mentally (or with simple notes) record their observations about the ball's movement on each ramp shape.
- **Critical Thinking:** Students will analyze their observations to determine which ramp shape seems best for a slow and safe descent.
- **Material Testing (If applicable):** Students will build ramps using swimming noodles, cardboard, and aluminum foil and compare how the toy wheelchair moves on each material.
- **Reflection:** Students will answer the reflection questions, explaining their findings based on their experiments.

Parts of the books included.





DIY Lab

Objective: Students will investigate how different ramp shapes and materials affect the speed, stability, and safety of movement for someone using a wheelchair. By designing and testing ramps with various materials, students will apply the principles of engineering, observation, and problem-solving to make thoughtful decisions that support inclusive and safe design.

Materials Needed:

- Cardboard pieces (for ramp base)
- Toy ball or marble (to simulate a wheelchair)
- Swimming noodles
- Aluminium foil
- Scissors, tape, and support blocks or books
- Images or models of different ramp types (spiral, straight, steep)

Activity Instructions

• Part 1: Test Ramp Shapes.

1. Review the 3 ramp shapes: spiral, straight, and steep.
2. Build each ramp shape using the same material (cardboard).
3. Roll a toy ball down each ramp and observe:
 - Did it roll fast or slow?
 - Did it bounce or roll smoothly?
 - Did it stay on the ramp safely?
4. Record your observations.
5. Pick your favorite ramp shape and explain why it worked best.

• Part 2: Test Ramp Materials.

1. Choose one ramp shape (e.g., spiral or straight).
2. Rebuild the same ramp using each of these materials:
 - Swimming noodle
 - Cardboard
 - Aluminium foil
3. Roll the toy ball down each material ramp and observe:
 - Which rolled the smoothest?
 - Which felt safest and most stable?
 - Which one was the strongest?
4. Record your answers in the reflection space provided.

To conclude the ramp investigation activity, bring students together for a short reflection and discussion to help them process what they discovered and reinforce key learning points.

“Today, you were real engineers! You tested different ramp shapes and materials to find out what helps Khaled—and others using wheelchairs—move down safely and smoothly. We learned that the design and the materials we choose really matter when we’re trying to help everyone play and move safely.”

To help Khaled use a wheelchair safely, the best ramp combination is:

Best Ramp Shape:

Spiral Ramp

Why:

- Offers a longer, gentler slope
- Slows down the descent naturally
- Provides continuous safety without steep drops
- Easier to manage for people using wheelchairs or mobility aids

Best Material:

Cardboard

Why:

- More stable than foil or noodles
- Doesn’t bend easily under weight
- Provides a smooth surface for rolling
- Easy to shape into ramps and support with blocks



Teacher Role:

- **Concept Explainer:** Clearly introduce the purpose of ramps in providing accessibility, particularly for individuals using wheelchairs or other mobility aids
- **Facilitator of Observation:** Guide students' viewing by highlighting key aspects of ramp design they should pay attention to, such as slope, surface material, presence of handrails, and overall configuration (straight vs. curved).
- **Prompter of Critical Thinking:** Encourage students to analyze the ramps shown in the video, prompting them to consider why certain design choices might be more effective or safer than others.
- **Discussion Moderator:** Lead a comprehensive post-viewing discussion, ensuring all students have the opportunity to share their observations, interpretations, and opinions about the ramps presented. Facilitate a deeper understanding of the principles of accessible design.

- **Synthesizer of Learning:** Help students connect the information from the video to the hands-on ramp building activity they will likely undertake, emphasizing how the video provides valuable insights for their own designs.

Expected From Students:

- **Active and Focused Viewing:** Students will watch the video attentively, concentrating on identifying the different features of the ramps shown and how individuals interact with them.
- **Detailed Observation:** Students will carefully observe the slope, surface, handrails (if present), and overall shape of the ramps in the video. They will also pay attention to how wheelchairs move on these different ramps.
- **Thoughtful Participation:** Students will actively contribute to the pre- and post-viewing discussions, sharing their observations, analyses, and insights gleaned from the video. They will also respond thoughtfully to the ideas and questions raised by the teacher and their peers.
- **Connection to Application:** Students will begin to make connections between the design principles observed in the video and the practical considerations for building their own ramps in subsequent activities.

Parts of the books included.



Screen Lessons

Watch the video to explore and learn something new about ramps together!

<https://www.youtube.com/watch?v=3COvm0TtxWg>



Screen Lessons

Objective: Students will understand the real-world importance of ramps by observing how ramps help people with different needs—such as those using wheelchairs, strollers, walkers, or delivery carts—move safely and independently. The video will help students build empathy, recognize design for accessibility, and see how engineering solutions like ramps make public spaces fair and usable for everyone.

Video link: <https://www.youtube.com/watch?v=3COvm0TtxWg>



Possible Strategies for implementation:

1. Pre-Watching: Set a Purpose.

- **Strategy:** What Do We Know? Before showing the video, ask students: "What are some things we already know about ramps? Where have you seen them? What makes a ramp easy or hard to use?" Write down their ideas. Explain that the video will help them learn more about what makes a good ramp.

2. During Watching: Active engagement.

- **Strategy:** Ramp Checklist: Before starting the video, give students a simple checklist of things to look for in the video (e.g., "Does the ramp have a gentle slope?", "Is the ramp straight or curved?", "What materials are used to build the ramp?", "Does it look easy or hard to use?"). They can check off the items as they see them in the video.

3. Post-Watching: Discussion and reflection.

- **Strategy 1:** "Good Ramp, Bad Ramp": After watching the video, discuss what made a ramp "good" (easy and safe to use) or "bad" (difficult or dangerous). Ask students to give examples from the video.
- **Strategy 2:** Design Your Own Ramp (Discussion): Based on what they learned from the video, ask students to describe their ideal ramp. What shape would it be? What material would it be made of? How would they make it safe and easy to use?
- **Strategy 3:** Draw a Ramp: Have students draw their ideal ramp, labeling the key features that make it accessible (e.g., "gentle slope," "smooth surface," "handrails").

Elaborate



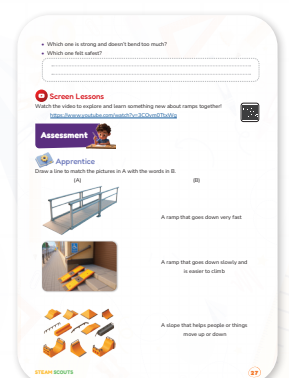
Teacher Role:

- **Clarifier:** Make sure students understand key vocabulary used to describe different ramp types (e.g., "goes down very fast," "goes down slowly," "easier to climb," "helps people move up or down").
- **Facilitator:** Provide students with materials such as cardboard squares and explain how to use them for layout planning.
- **Supporter:** Guide students in numbering and arranging the playground elements, helping them connect their digital Tinkercad design to a physical layout.
- **Observer:** Watch how students match descriptions to images, use spatial reasoning, and apply planning strategies to structure their playground sequence.

Expected From Students:

- **Matching & Reasoning:** Students will match ramp descriptions to the correct images by thinking about how each ramp functions in real life.
- **Planning & Layout:** Students will use numbered cardboard squares to represent sections of their playground, translating their digital design into a real-world arrangement.
- **Numbering & Sequencing:** Students will number the layout areas from 1 to 12 and assign locations to specific playground elements, thinking critically about the order and structure of their design.

Parts of the books included.





Apprentice

Objective:

Students will demonstrate their understanding of different ramp types by matching visual images of ramps with their correct functional descriptions. This activity helps build vocabulary, develop visual reasoning, and reinforce the concept of how slope and design affect safety and usability.



Apprentice

Draw a line to match the pictures in A with the words in B.

(A)

(B)



A ramp that goes down very fast



A ramp that goes down slowly and is easier to climb



A slope that helps people or things move up or down

STEAM SCOUTS

27

Possible Strategies for Implementation:

- **Observe the Slope:** Look closely at how steep each ramp appears. A less steep ramp will result in a slower descent and easier climb.
- **Consider Handrails:** Handrails indicate a ramp designed for easier use by people, suggesting a slower and more accessible design.
- **Think About Function:** Ramps are designed to help things move up or down, so this description applies to most of the images.



Builder

Objective: Students will translate their digital Tinkercad playground design into a physical model layout using a grid system. They will use number lines, counting, and ordinal numbers to plan and sequence the placement of playground elements. This activity builds skills in spatial reasoning, organization, math, and design planning.

Materials Needed

- Cardboard or printed paper grid board (12 numbered squares)
- Markers or pencils
- Copy of student's Tinkercad playground design (digital or printed)

Activity Instructions

- **Introduce the Purpose.**

Explain to students that they'll now take their digital playground plan and prepare to "build it" using a grid board, just like a construction team would.

Emphasize that this activity helps them think about what comes first, where things go, and how to organize their ideas.

- **Distribute Materials.**

Hand out the 12-square grid boards and markers.

Ensure students have access to their Tinkercad design for reference.

- **Guide the Numbering (if needed).**

Help students number their squares from 1 to 12.

You can project a number line or model it on the board.

- **Assign Spaces to Playground Elements.**

Prompt students to count how many of each element they have in their digital design (e.g., swings, slides, ramps).

Ask them to assign a square to each one by writing its name or using small markers.

- **Encourage Sequencing Using Ordinal Terms.**

Invite students to think:

"What will you build first?"

"What comes next?"

"What should go last to make sure everything fits and works well?"

Let them write it down in the book.

- **Facilitate Sharing.**

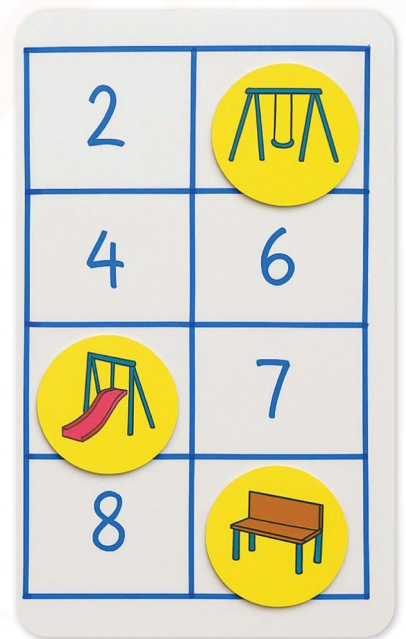
Pair students or do a gallery walk. Ask guiding questions:

- "Why did you put your ramp first?"

- "How does your layout help all kids play safely?"

Teacher Role:

- **Facilitator:** Guide students in the initial construction of their physical playgrounds using the cardboard squares as a base.
- **Observer:** Observe students as they follow the instructions to add the spiral ramp road, green area, and Tinkercad accessories. Note their ability to translate their digital design into a physical layout.
- **Supporter:** Offer assistance with spatial reasoning, connecting the Tinkercad design to the physical space, and basic construction techniques (if applicable in this initial stage).

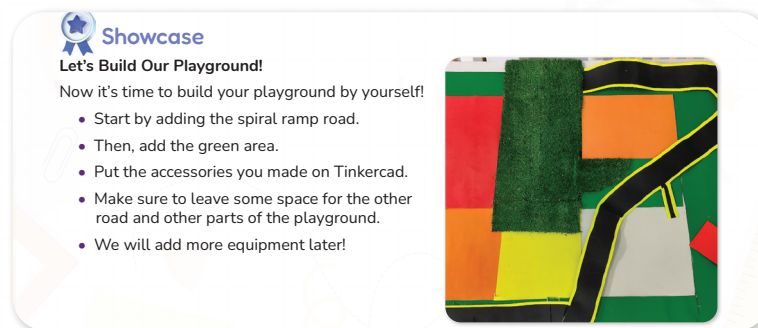


- **Encourager:** Encourage students to think about leaving space for future additions and the overall flow of their playground.

Expected From Students:

- **Following Instructions:** Students will follow the initial steps to begin building their physical playground.
- **Spatial Reasoning:** Students will attempt to translate the spiral ramp road and green area from their Tinkercad design onto the cardboard square base.
- **Integration of Tinkercad Elements:** Students will consider how the "accessories" they created in Tinkercad will eventually be represented in their physical model.
- **Planning for Future:** Students will demonstrate an understanding of leaving space for other playground elements.

Parts of the books included.



Showcase

Objective: Students will apply their understanding of playground planning by assembling a 3D physical model using a paper spiral ramp, colored paper, and their digital Tinkercad components. This hands-on activity develops spatial awareness, fine motor skills, and design application by transforming digital concepts into a real-world visual model.

Activity resources: <https://pin.it/50ObGX9J9>



Materials Needed:

- Printed paper spiral ramp template (cuttable)
- Colored Canson paper for green spaces and accessories
- Cardboard base (size depends on class setup)
- Glue sticks or liquid glue
- Scissors
- Student's printed Tinkercad playground accessory designs (optional)
- Markers for decorating paths, labeling features, or adding details

Activity Instructions:

- **Review the Goal.**

Remind students that their task is to bring their playground to life. Start with the spiral ramp, then add green zones and other features.

- **Cut the Spiral Ramp.**

Distribute the pre-printed spiral ramp template. Instruct students to carefully cut along the lines. Support fine motor skill needs as needed.

- **Glue the Ramp.**

Guide students to gently twist and form the ramp, then glue it in a spiral shape onto the cardboard base.

- **Add Green Space.**

Let students cut shapes from yellow Canson paper and glue them around the spiral as parks or soft areas.

- **Decorate with Tinkercad Accessories.**

Add students printed Tinkercad pieces.

- **Leave Space for More.**

Remind students to leave space for more features to be added in the next phase (e.g., roads, ramps, rest areas).

- **Display and Share.**

Possible Strategies for implementation:

1. Model the Process First (I Do – We Do – You Do)

- Start by demonstrating how to cut and form the spiral ramp using a large model in front of the class.
- Then do a short guided practice with students before allowing independent building.
Helps reduce confusion and builds confidence with tricky steps like spiraling and gluing.

2. Use a "Ramp Station" Setup

- Set up the classroom with building stations:
 - Cutting Station (spiral templates)
 - Coloring Station (green area cutouts)
 - Assembly Station (cardboard and gluing)
- Keeps the process organized and supports differentiated pacing.

3. Visual Examples for Reference

- Show photos of real spiral ramps or sample models on the board or display.
- Let students compare their models to these for inspiration.

Supports visual learners and provides real-world connections.

4. Use Peer Buddies or Team Builds

- Pair or group students to work together on one model (if time or materials are limited).
- Assign roles like cutter, gluer, designer, or decorator.
Encourages collaboration and distributes responsibilities.

5. Incorporate Vocabulary Review During Work

- While students build, ask guiding questions:

“Is your ramp curved or steep?”

“What shape are you using for the green area?”

Reinforces key language and concept connections during hands-on learning.

6. Celebrate & Reflect

- After models are completed, host a “Playground Walkthrough” where students present:

“Here is my spiral ramp... This green area is for resting... This part came from my Tinkercad design...”

Showcase: Rubric for Evaluating Initial Physical Playground Construction

Criteria	Beginning (1 Point)	Developing (2 Points)	Proficient (3 Points)
Spiral Ramp Road (Initial)	Shows limited attempt to represent a spiral ramp road on the base. Shape is unclear or significantly different from the Tinkercad design.	Shows some attempt to create a spiral ramp road. Shape is recognizable as a spiral but may lack detail or accurate scaling.	Clearly represents a spiral ramp road on the base, showing a reasonable connection to the Tinkercad design in terms of shape and placement.
Green Area (Initial)	Green area is missing or very vaguely defined. Placement does not reflect the Tinkercad design.	Shows some designated green space. Placement has a basic connection to the Tinkercad design.	Clearly indicates a green area on the base, with placement that aligns with the Tinkercad design.
Consideration of Tinkercad Accessories	No clear indication of where Tinkercad accessories might be placed.	Shows some initial thought about including Tinkercad accessories in the physical space.	Demonstrates a clear plan for incorporating Tinkercad accessories into the physical layout, considering their size and function.
Space Planning (Future Elements)	No space is left for other playground parts, or the arrangement seems haphazard.	Shows some awareness of needing space for other elements, but the spacing may not be well-planned.	Demonstrates thoughtful planning by leaving appropriate space for other roads and playground parts, considering the overall layout.
Following Instructions	Struggles to follow the initial building instructions.	Follows most of the initial building instructions with some prompting.	Follows the initial building instructions accurately and independently.

Don't forget to go back to the EDP page and add these to planning part.



Now I Can

Objective: Students will reflect on and self-assess their learning and skills gained throughout the spiral ramp and playground design project. By reviewing and checking off "I Can" statements, they demonstrate their understanding of key STEAM concepts such as designing inclusive play areas, material testing, model making, and math-based planning.

Activity Instructions

1. Introduce the Reflection:

Let students know they are going to think back on everything they've done in this lesson.

2. open "Now I Can" page in the book:

ask the students to open now I can page in the book.

3. Read and Review Together:

Read each statement aloud and briefly remind students what that part of the project looked like (e.g., "Remember when we tested materials like foil and cardboard?")

4. Student Check-In:

Ask students to place a check mark next to each statement they feel confident they completed or understood.

Relation to Standards:

1. Next Generation Science Standards (NGSS):

- **K-2-ETS1-2:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem. (Students are building a physical model of their functional playground design.)
- **K-2-ETS1-3:** Analyze data from tests of two objects designed to solve the same

problem to compare the strengths and weaknesses of how each performs. (As students build and potentially modify their designs, they are implicitly engaging in this process.)

2. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening):

- **SL.1.5:** Add drawings or other visual displays to descriptions when appropriate to clarify ideas, thoughts, and feelings. (The physical model serves as a visual display of their Tinkercad design.)

3. Common Core State Standards (CCSS) - Mathematics (Geometry):

- **1.G.A.2:** Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, half-circles, and quarter-circles) or three-dimensional shapes (cubes, right rectangular prisms, right circular cones, and right circular cylinders) to create a composite shape, and compose new shapes from the composite shapes. (Students are using their understanding of shapes to construct their playground elements.)